EVERY ASSOCIATED PRIME OF A MODULE LIES IN ITS SUPPORT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Associated primes via annihilators of elements. Used at the first line of the counted proof (line 15341). Provides an element m whose annihilator is the given prime, which is the witness the proof localises.
- Nonvanishing of a localisation as a support condition. Used at the end of the counted proof (lines 15343-15344). Turns the class of m being nonzero in the localisation at p into membership of p in the support.

Lemma 0.1. *Let R be a ring. Let M be an R-module. Then $Ass(M) \subset Supp(M)$.*

Proof. If $m \in M$ has annihilator $\mathfrak{p}$, then in particular no element of $R \setminus \mathfrak{p}$ annihilates m. Hence m is a nonzero element of $M_{\mathfrak{p}}$, i.e., $\mathfrak{p} \in \mathrm{Supp}(M)$. □

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